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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-SE-018

1. Topic (12 words max)

AI Powered Smart Orchid Care System Using Multi-Modal Machine Learning

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Vanda orchid cultivation is a high-value floriculture activity practiced in controlled greenhouse and nursery environments. However, orchid care remains largely dependent on manual observation and grower experience. Commercial floriculturists, nursery managers, and research institutions are affected by challenges such as delayed disease detection, improper watering schedules, inaccurate nutrient application, and difficulty in predicting growth stages and flowering periods. Due to the epiphytic root structure and high sensitivity of Vanda orchids to environmental changes, these issues result in reduced flower quality, increased plant losses, and higher production costs, particularly for small- and medium-scale growers [1], [3].

Impact of the Problem

- Economic Impact:
  - Commercial Vanda orchid producers face financial losses caused by fungal diseases and rare disease outbreaks that spread rapidly in greenhouse environments. Delayed detection often requires discarding infected plant batches, negatively affecting yield consistency and export revenue [1], [4].
- Technical Impact:
  - Conventional irrigation and nutrient management practices lack precision. Over-watering leads to root rot, while under-watering restricts nutrient absorption. Existing systems do not adapt water and nutrient supply to different growth stages, resulting in inefficient resource usage and suboptimal plant development [2], [5].

- Environmental Impact:
    - Inefficient irrigation and excessive use of chemical fertilizers and fungicides increase water wastage and environmental pollution, raising the ecological footprint of orchid cultivation [1], [6].

FoC Research Cluster and Industry Verticals

- Research Cluster: Artificial Intelligence (AI), Internet of Things (IoT), and Multi-Modal Machine Learning
- Industry Vertical: Smart Agriculture (Precision Farming) and Floriculture

Sustainable Development Goals (SDGs)

- This research contributes to SDG 6 (Clean Water and Sanitation) through precision irrigation and water forecasting models that reduce unnecessary water consumption [2], [6]. Furthermore, the project aligns with SDG 12 (Responsible Consumption and Production) by optimizing fertilizer application [1], [5].

References

[1] D. Wijendra et al., “OrchiZen: Hybrid Integrated Smart Farming System for Orchid Plantations,” Proceedings of the 2nd International Conference on New Frontiers in Communication, Automation, Management and Security (ICCAMS), 2025.

[2] R. P. G. S. Maleesha and P. D. S. Silva, “Analyzing the Influence of Automated Water Distribution Systems on Precision Irrigation for Orchids: A Case Study Using Dendrobium–Phalaenopsis Orchid Group,” 2024.

[3] C. Y. T. Chen et al., “CWT IoT Device for Detecting Rare Events of Orchid Disease,” IEEE Internet of Things Journal, vol. 11, no. 12, pp. 22830–22842, Jun. 2024.

[4] L.-W. Wu, S.-T. Chen, and Y.-C. Ouyang, “Detection of Phalaenopsis Fusarium Wilt from Hyperspectral VNIR/SWIR Imagery Using EE-CSPNet,” 2024.

[5] A. Sungheetha et al., “Multi-Modal Optical Sensing Fusion for Quantitative Monitoring of Orchid Parameters: Analysis of Commercial Cultivation in Asia,” Proceedings of the 2nd International Conference on New Frontiers in Communication, Automation, Management and Security (ICCAMS), 2025.

[6] Sukarwoto et al., “Internet of Things on Automatic Watering Systems for Papuan Black Orchids,” 2023.

[7] Y.-B. Lin et al., “Design of a Spore Germination Sensor for Orchids,” IEEE Sensors Letters, vol. 9, no. 2, Feb. 2025.

6. Brief description of Existing Research and Systems (300 words max)

Existing research and commercial systems related to orchid care mainly focus on using artificial intelligence and smart technologies to improve plant health, watering, and disease management. Several studies and products show that these technologies can help increase orchid yield and reduce resource waste. However, most existing solutions are either hardware-heavy, limited to a single function, or not fully integrated.

Summaries of existing products and research:

- Orchid Care – Plant Identifier allows users to identify orchid varieties using a camera, diagnose visible health issues, track watering routines, and chat with an AI-based orchid assistant.
- OrchiZen combines image-based disease detection with environmental monitoring to automate irrigation, lighting, and fertilizer advice.
- ChatbotPowered_Orchid_Leaf_Disease_Classification_and_Management_for_Improved _Farming have also been introduced, allowing farmers to upload leaf images and receive disease classifications through mobile platforms.
- Smart watering and nutrient forecasting, many research studies use IoT sensors, machine learning, reinforcement learning, and digital twin models to control irrigation and fertilizer application. These systems improve efficiency and accuracy but often require costly sensors, complex calibration, and regular maintenance, making them less practical for small-scale or home growers.
- Research on orchid growth monitoring uses multi-modal optical sensing techniques, such as hyperspectral imaging and fluorescence analysis. While highly accurate, these methods are expensive and mainly suitable for large commercial greenhouses.

Gaps / limitations identified:

- Orchid disease detection, Severity assessment and Treatment recommendation –
  Existing systems mainly detect the presence of diseases but do not assess severity, predict disease progression, or recommend treatments based on disease type and severity in orchids.
- Smart watering and nutrient forecasting –
  Most smart watering and growth monitoring solutions rely heavily on IoT sensors and expensive equipment, making them unsuitable for small-scale growers. Also detect nutrient status but do not forecast nutrient dosage or support automated nutrient supply.
- Orchid growth stage recognition and bloom-date prediction –
  There isn't post bloom monitoring and we are taking vanda as the species.
- Hybrid pollination and Comparability Analysis –
  Current research and commercial products do not use machine learning for orchid hybrid pollination or compatibility prediction. Pollination decisions remain manual and experience-based without data-driven support.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is an **AI-powered Smart Orchid Care System** that integrates IoT sensors, computer vision, and multi-modal machine learning to support the complete orchid cultivation lifecycle. The system is designed to overcome the limitations of existing orchid care solutions by combining disease intelligence, smart watering and nutrient forecasting, growth-stage prediction, and hybrid pollination analysis into a single, unified platform.

Existing systems mainly focus on separated functions such as disease detection or watering automation. They do not assess disease severity, nutrient forecasting, blooming timelines, or support data-driven hybrid pollination decisions. Our solution fills these gaps by:

- Assessing disease severity and progression instead of simple detection.
- Forecasting water and nutrient requirements using both sensor data and visual plant traits.
- Predicting growth stages and bloom dates for Vanda orchid species.
- Introducing AI-based hybrid pollination and compatibility analysis.

System Perspective of the Solution: Inputs, Process, and Outputs

Inputs

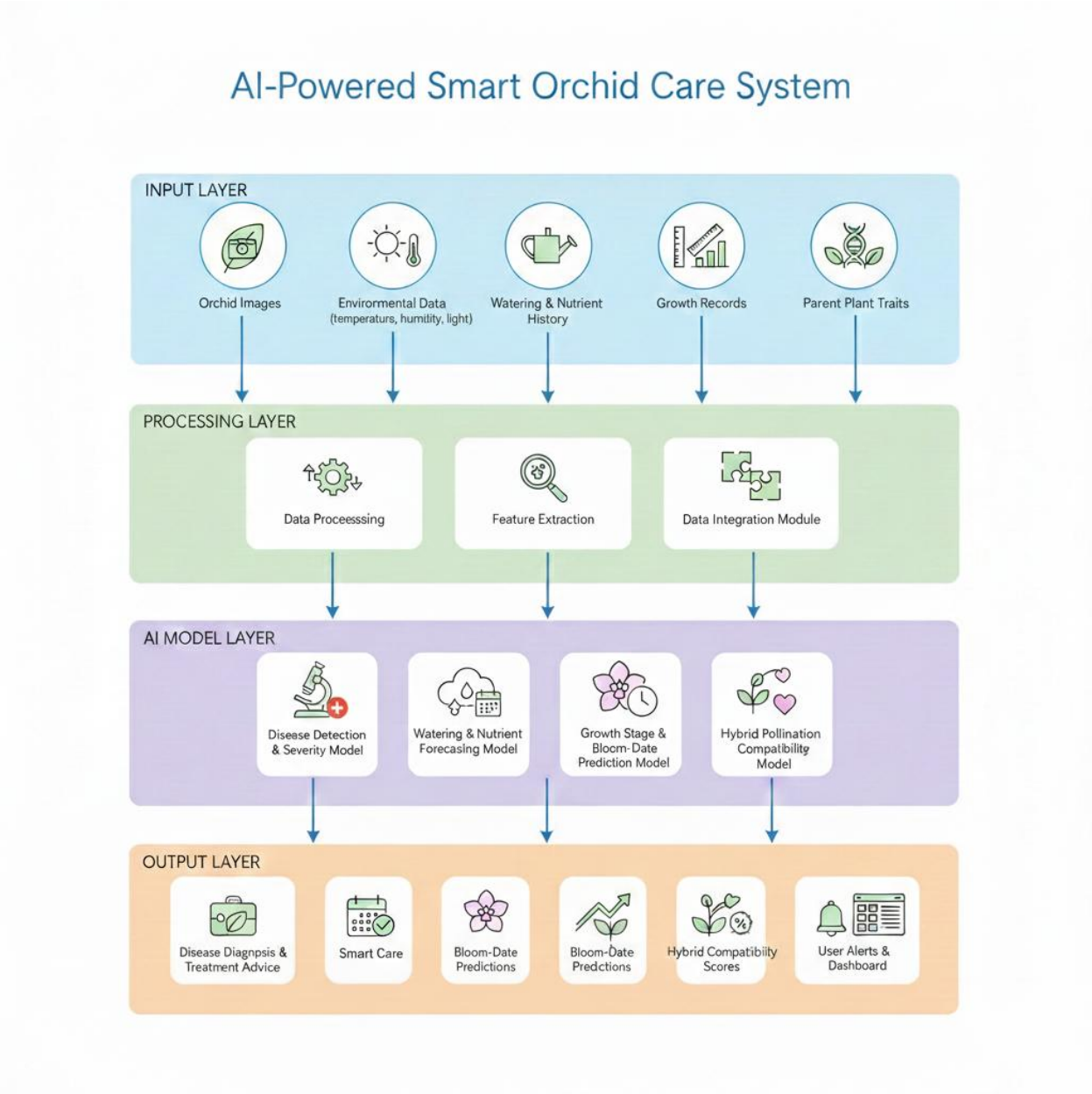
- Real-time sensor data including temperature, humidity, light intensity, and moisture levels from the orchid environment.
- High-resolution images of orchid leaves, stems, and flowers captured periodically.
- Historical orchid growth records and flowering timelines.
- Past disease images and treatment outcomes.
- Hybrid parent orchid characteristics and pollination history.

Process

- The system studies orchid images to find diseases, understand how serious they are, and suggest suitable treatments according to severity level of the disease.
- Uses sensor readings and plant appearance to decide how much water and nutrients the orchid will need.
- Observes orchid growth patterns to identify its current stage and predict when it will bloom.
- Compares parent orchids to decide which combinations are most likely to produce successful hybrids.

Outputs

- Detailed disease severity reports with stage-wise treatment guidance.
- Optimized water and nutrient forecasts for upcoming growth cycles.
- Accurate growth stage identification and bloom-date predictions.
- Hybrid pollination compatibility scores and breeding recommendations.
- Actionable insights delivered through a unified smart orchid care platform.


AI-Powered Smart Orchid Care System
INPUT LAYER
Orchid Images
Enviromental Data (temperatus, humitity, light)
Watering & Nutrient History
Growth Records
Parent Plant Traits
PROCESSING LAYER
Data Proceesssing
Feature Extraction
Data Integration Module
AI MODEL LAYER
Disease Detection & Severity Model
Watering & Nutrient Forecasing Model
Growth Stage & Bloom-Date Prediction Model
Hybrid Pollination Compatibility Model
OUTPUT LAYER
Disease Diagnpsis & Treatment Advice
Smart Care
Bloom-Date Predictions
Bloom-Date Predctions
Hybrid Compatibiity Scores
User Alerts & Dashboard

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Successful implementation of the AI Powered Smart Orchid Care System for Vanda species requires interdisciplinary expertise in orchid biology, plant pathology, artificial intelligence, smart agriculture, and data management. The development of the necessary expertise, technical knowledge, data, and ethical compliance will be done as follows.

Acquisition of specific domain expertise

Essential domain knowledge related to Vanda orchid cultivation, physiology, and pathology will be acquired through a structured review of scientific literature, orchid cultivation manuals, and agricultural extension publications focusing on Vanda species. The project team will consult with subject matter experts such as horticulturists, orchid nursery operators, and academic staff from the Departments of Agriculture and Plant Sciences to validate disease symptoms, growth stage definitions, pollination practices, and treatment methods. Participation in orchid research forums and engagement with local commercial Vanda growers will further strengthen practical understanding and real-world relevance.

Technical knowledge and skills development

Technical skills required for this project include computer vision, time series forecasting, multi-modal machine learning, and IoT-based data acquisition. These skills will be developed through self-learning using established machine learning frameworks, and practical experiments on convolutional neural networks (CNNs) for disease detection and growth-stage recognition will also be conducted. Practical experience with environmental sensors (temperature, humidity, light intensity, and moisture exposure) and cloud-based analytics platforms will be gained through prototype development and iterative testing. Open-source research implementations and published IEEE studies will be used as technical references.

Data Availability and Collection

Primary data will be collected through controlled observation of Vanda orchids grown in nurseries, shade houses, or greenhouse environments. This includes high-resolution images of leaves, stems, and flowers under healthy and diseased conditions. Environmental data such as temperature, humidity, light intensity, watering intervals, and nutrient application records will be collected using IoT sensors. Disease type, severity level, growth stage, and treatment results will be reported with expert validation. In addition, hybrid pollination data including parental combinations, interbreeding success rates, and flowering performance are reported to support the models for compatibility analysis.

Secondary data are obtained from publicly available plant disease image datasets, orchid research publications, agricultural databases, and horticultural extension guidelines to supplement training and improve model robustness.

Ethical considerations

The proposed project does not involve human subjects, animals, or sensitive personal data and therefore does not require formal institutional ethical clearance. However, ethical research practices are strictly followed. Consent is obtained from orchid growers before plant or environmental data are collected, farm identification information is anonymized, and all data are securely stored and used for study purposes only. Human-in-the-loop validation is maintained to prevent unsafe automated treatment recommendations.

9. Objectives and Novelty

Main Objective

The main objective of this project is to develop an AI-powered smart care system for Using Multi-Modal Machine Learning *Vanda* orchids that integrates disease detection, growth-stage recognition, bloom-date and flower longevity prediction, smart watering and nutrient forecasting, and hybrid pollination compatibility analysis. The system uses multi-modal data (images, sensors, and growth history) to provide predictive care, accurate bloom forecasting, and support for hybridization planning.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Wickramasinghe D.P IT22065230	Hybrid pollination and compatibility Analysis	•Study and document the principles of hybrid pollination and compatibility factors in orchid species, focusing on the Vanda orchid. •Collect and organize parent plant characteristics, blooming qualities, and history hybridization data from studies and professional input. •Create a machine learning model to analyze parental orchid traits (species traits, bloom time, flower structure) for compatibility prediction. •Develop a compatibility scoring mechanism to rank potential hybrid parent pairs based on predicted success probability.	•This component introduces ML-based hybrid compatibility prediction for orchids. •Focuses on Vanda orchid hybridization, which is underrepresented in current research compared to Phalaenopsis. •Introduces a quantitative compatibility score to guide breeders in selecting parent orchids before pollination. •Bridges the gap between orchid biology and AI by transforming subjective pollination experience into a predictive system.

Rathnamalala D M J P IT22064936	Orchid disease detection, Severity assessment and Treatment recommendation	• Identify common orchid diseases affecting leaves, roots, and flowers, including fungal, bacterial, and environmental stress-related issues. • Collect and prepare orchid images showing healthy plants and different levels of disease severity. • Develop an image-based model to detect orchid diseases and classify their severity as early, moderate, or severe. • Combine environmental conditions such as humidity, temperature, and light with image findings to improve disease assessment accuracy. • Generate suitable treatment recommendations based on the identified disease type and severity.	• Extends simple disease detection by assessing severity and progression stages in orchids. • Combines visual symptoms and environmental conditions to improve disease understanding and accuracy. • Introduces an AI-driven treatment recommendation approach according to severity level. • Designed specifically for Vanda orchids, addressing species-specific disease behavior rather than using generalized plant models. • Provides early warning insights, enabling preventive measures before irreversible damage occurs.
Vithanage P V P M IT22062642	Smart watering and nutrient forecasting	• Study water and nutrient requirements of Vanda orchids at different growth stages. • Identify key environmental and substrate parameters affecting water and nutrient uptake (temperature, humidity, light, substrate moisture, EC/pH). • Collect and preprocess multi-modal time-series data from sensors and historical growth records.	• Introduces forecast-based smart watering for Vanda orchids, predicting future water needs instead of using fixed moisture thresholds. • Implements automated fertilizing for Vanda orchids, supplying nutrients automatically rather than only giving fertilizer recommendations. • Integrates intelligent fertigation, combining water prediction and

		• Develop a machine learning model to forecast future water requirements (when and how much to irrigate). • Develop a nutrient demand prediction model to estimate nutrient dose and timing based on plant stage and environmental conditions. • Integrate watering and nutrient forecasting into a smart fertigation recommendation module. • Validate predictions using water usage records, nutrient application logs, and plant health indicators.	automated fertilizer supply in a single control module. • Integrates automated fertigation logic by combining water and nutrient prediction in a single intelligent module, enabling coordinated irrigation and nutrient supply tailored to the plant's needs. • Focuses specifically on Vanda orchids, an orchid species that is underrepresented in existing smart irrigation and nutrient management research, and whose epiphytic nature requires precise water and nutrient control.
Nimla Samadinee L.D IT22083500	Orchid growth stage recognition and bloom-date prediction	• Study and define distinct growth stages of Vanda orchids (juvenile, vegetative, spike initiation, bud development, flowering, post-flowering). • Collect and preprocess images of Vanda orchids across different growth stages. • Develop a computer vision model for automatic growth stage classification. Integrate environmental and historical growth data (temperature, humidity,	• Unlike existing studies that focus on Phalaenopsis, this growth-stage recognition model is specifically for Vanda orchids. • Integration of growth-stage recognition with flower maturity prediction to estimate how long a bloom will last. • Use of multi-modal data fusion (imagery + environmental data) for more accurate bloom date and longevity prediction.

		light, watering) with image-based features. • Implement time series models to predict bloom dates and flower lifespan. • Validate predictions using actual bloom date and flower shedding records.	• Practical relevance to growers for planning flowering cycles and hybridization schedules.

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Topic is accepted.
Suggested to fine tune the novelty and plan the dataset collection.

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Hansi	DeSilva	
Co-Supervisor	Ms	Thilini	Jayalath	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				